

Welcome to **instats**

The session will begin shortly.

Start

THIRD MOVEMENT • 35 MINUTES

Crawling, with manners.

The two-screen check that decides whether you can crawl a site at all. The five-line Python that does the actual work.

03

Read robots.txt before any code runs.

```
# https://www.csis.org/robots.txt User-agent: * Allow:  
/analysis/ Allow: /*?page= Disallow: /admin/ Disallow: /search/  
Disallow: /user/ Disallow: /comment/ User-agent: Bytespider  
Disallow: / # Verified 2026-04-28. Re-check # before any second-  
pass crawl.
```

THREE THINGS TO FIND

- **User-agent.** Which crawlers each rule block applies to.
- **Allow.** Paths you may fetch.
- **Disallow.** Paths you may not fetch.

WHY IT MATTERS BEFORE ANY CODE

Robots.txt is a social contract. It does not have legal force in most jurisdictions. It is the first thing a methods reviewer will ask whether you checked. Document the date you read it. Save a copy. Re-check before any second pass.

The Python is short. The hygiene is the whole job.

```
# A polite fetch in 8 lines.
import requests, time

UA = 'workshop-demo (mailto:you@univ.edu)'

def polite_fetch(url):
    r = requests.get(url,
        headers={'User-Agent': UA},
        timeout=10)
    if r.status_code != 200:
        return None
    time.sleep(1.0)
    return r.text # raw HTML
```

- **User-Agent.** A real string that identifies you and gives a contact email. Sysadmins should be able to reach you instead of blocking your IP.
- **Status check.** 200 is the only happy path. 403 means blocked. 404 means moved. 429 means slow down. 503 means try later.
- **Timeout.** Default 10 seconds. Without it your script hangs on a slow page and never tells you why.
- **Raw HTML.** Store the bytes, not just the parsed text. Slide 21 explains why.
- **Sleep.** One second between requests is the floor. Slide 22 details politeness limits.

Crawl one CSIS analysis page together.

What we run

Section 9. Define `polite_fetch()`. User-Agent string. Timeout. Status check. Returns raw HTML and metadata.

Section 10. Fetch one article URL. Parse with BeautifulSoup. Pull the headline, publication date, and first paragraphs. Inspect the parsed output. Note the raw HTML length, around 100 KB.

CSIS is server-rendered HTML. No JavaScript execution required. That is why generic Python crawls work here, and why the same approach can fail elsewhere. If your live test fails, the site is telling you to use their public API.

Save the raw HTML after every fetch.

REASON 01

Parsers change

You will discover next month that the date selector also matches a sidebar element. Re-running the parser on stored HTML is a one-line script. Re-crawling the site is a one-day script and a fresh robots check.

REASON 02

Sites change

The article you scraped today may not look the same in six months. Stored raw HTML is the only durable evidence that your data existed in the form you claim.

REASON 03

Replication packages

You ship the parser plus the raw HTML. A reviewer running your code in 2030 should not need network access to reproduce your dataset.

Filename is the URL slug plus a timestamp. Compress as gzip if the corpus grows.

Be slower than you think you need to be.

RATES OF FIRE

PACE	VERDICT
0.5 / sec	Polite. Default for production crawls.
1 / sec	Floor. Today's notebook setting.
5 / sec	Rude on most sites. Avoid.
20 / sec	Denial of service. Never.

THREE IMPLEMENTATION DETAILS

- **Sleep after each request** rather than before the first. The point is to space the load on the server. Spacing your own script is a side effect.
- **Cache responses by URL hash.** If you re-run, do not re-fetch what you already have.
- **Log every request** with a timestamp. If a sysadmin emails you, you can show your traffic pattern from your own logs.

CSIS does not publish explicit rate limits. The default in the demo is one second between requests. Tune slower if your IP starts catching 429 responses. Never speed up.

FOURTH MOVEMENT • 25 MINUTES

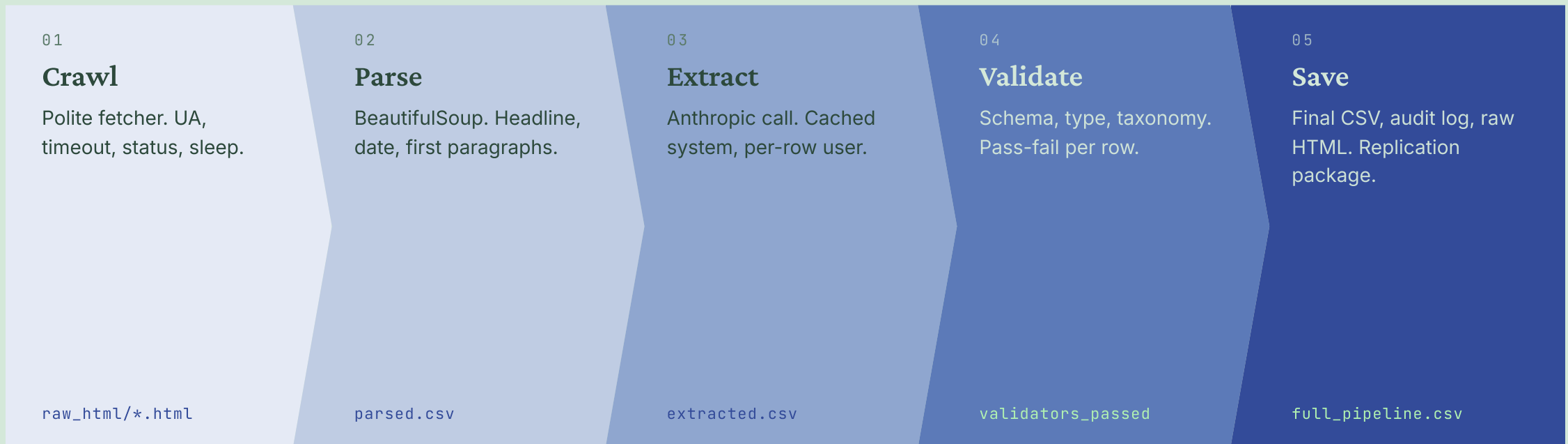
End-to-end pipeline.

Crawl, parse, extract, validate, save. Run together on five fresh articles.

04

FIVE STEPS, EACH ONE WRITES TO DISK

Architecture decision. Every step is restartable.



Each step reads from disk, writes to disk. If extract fails, you do not re-crawl. You re-extract.

Five fresh CSIS articles through all five steps.

What we run

Section 11. Five article URLs from the analysis index. Crawl with one-second sleeps. Save raw HTML. Print the manifest.

Section 12. Parse each raw HTML with BeautifulSoup. Pull headline, date, first three paragraphs. Save the cleaned CSV.

Section 13. Run the extractor on each parsed row. Cached system prompt. Per-row user message. Save extraction results next to the parsed text.

Section 14. Run validators. Print the pass rate. Print one full row from URL through validators-passed for inspection.

If any URL has changed since the workshop was prepared, switch to cached fallback HTML in the workshop GitHub repo at `day3/cache/`.

What gets saved. Twelve columns. One row.

```
# One full row, all columns
url: https://www.csis.org/...
fetched_at: 2026-04-28T09:14:22Z
status: 200
raw_path: raw_html/ai-and-future...html
headline: "AI and the Future of Mediation"
date_text: Apr 27, 2026
n_paragraphs: 14
extracted_json: {"signatories": ["UN", "WTO"],
  "agreement_type": "technology",
  "year": 2026,
  "sector": "technology"}
model_id: "claude-haiku-4-5-20251001"
system_prompt_hash: a1b2c3d4...
input_tokens: 412
output_tokens: 48
validators_passed: True
```

WHY THIS ROW

A reviewer can take this row and trace it back to a fetched HTML file, to a parser version, to a model version, to an exact prompt, to a token cost.

TWELVE MONTHS FROM NOW

You will thank the past version of you who saved every column. The cost of saving more is zero. The cost of saving less is a request for the appendix you did not write.

Provenance for one row equals provenance for the dataset. The pipeline either saves it for every row or it saves nothing usable.

CLOSING • 20 MINUTES

Reproducibility.

The appendix your reviewer is looking for. The arc you just completed. What to do this week.

05

The reproducibility appendix for an LLM-based measurement paper.

MUST-HAVE

- The full prompt verbatim, system and user, with date finalized.
- Model ID with version pin. Today's pin is `claude-haiku-4-5-20251001`.
- Temperature and any other generation parameters set.
- Schema with named taxonomies.
- Schema-validation pass rate on the full corpus.
- Robots.txt at date of crawl, plus your User-Agent string.
- Raw-HTML files in the replication package, or a stable archive mirror.
- Audit log with one row per call.

OFTEN ASKED FOR

- A small hand-coded gold set with field-level F1 against the model.
- A determinism rate from running the pipeline twice.
- Subgroup pass-rate breakdown. Sector, region, length.
- Total cost summary at production scale.
- Simpler-baseline comparison. Rule-based or smaller model.

If column one is in the appendix, the pipeline is defensible. Column two is where revisions go.

The three-day arc.

DAY 01 • THE MODEL ZOO

Two families, one decision

You can name the encoder and generative families and articulate the choice. You ran a state-of-the-art encoder on real political text in your browser.

DAY 02 • THE MEASUREMENT

From notebook to proxy

A codebook, a thirty-paragraph gold set, a thirteen-hypothesis ensemble, a country-year proxy plotted against external events, a validation appendix.

DAY 03 • THE PIPELINE

Wrangling and crawling

A structured extraction pipeline, a polite crawler, a five-step end-to-end pipeline, an audit log that ships with the paper.

The encoder and the generative model do different jobs in the same project. The hard part of measurement work in 2026 is the codebook, the gold set, the validation, and the audit. Those are the same on both sides. The model just changes which knob you turn.

Three days. Two families. One audit log.

THREE THINGS TO DO THIS WEEK

- Re-run the Day 3 notebook on five articles of your own choosing.
- Write a one-page replication appendix using slide 28 as the template.
- Pick the next concept. Spend an hour with a colleague writing the codebook.

OPTIONAL READING

Grimmer, Roberts, and Stewart, *Text as Data* (Princeton 2022). The reference text for measurement work in our field. Validation chapters map directly onto the appendix you just learned to ship.

Replication package. github.com/jacqpark/instats-llm-workshop

Thanks for spending three days here.

Stop